Fake Faces, Real Bias: A Multi-Ethnic Benchmark for Deepfake Detection

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Abstract

Deepfake detectors are gatekeepers of digital identity. Yet whether they guard all humans equally remains an open question. We introduce FairFaceBench, a demographically controlled video dataset built on the FHIBE dataset, spanning nine world regions, two genders, and seven generation methods including state-of-the-art avatar synthesis systems (Gemini Veo 3.1, Sora 2), with verified nationality and gender metadata. We evaluate four detectors (DINOv2 by Synthetiq Vision, Xception, EfficientNetB4, and F3Net from DeepfakeBench) and analyze bias from two complementary perspectives: false-alarm disparities on real faces, which cause legitimate users to be wrongly denied service and miss-rate disparities on synthetic faces, which allow demographically targeted fraud to bypass detection. Our analysis reveals that (i) the newest generative avatar models evade all tested detectors, and (ii) all four tested detectors exhibit substantial demographic bias, with worst-group AUC below 52% and mean subgroup AUC deviation up to 9.7 pp for zero-shot baselines. To address these disparities, we fine-tune all four detectors on FairFaceBench and show that overall AUC improves by 19-26 pp and worst-group AUC by 25-42 pp. Fine-tuning incurs an expected distribution-shift penalty on FF++ in-distribution data (7-24 pp AUC drop), which is partially mitigated by a worst-group loss objective that additionally improves fairness across generation categories.

1. Introduction

Digital identity verification has become a cornerstone of modern financial services, e-government, and online platforms. Regulatory requirements such as Anti-Money Laundering directives and Know-Your-Customer (KYC) mandates increasingly demand that customers present a live video selfie or participate in a real-time video call as proof of identity [8]. Synthetic face videos generated by modern AI systems can impersonate real individuals with sufficient realism to fool human reviewers and automated verification pipelines alike.

A primary attack vector targets video-based KYC sessions directly: financial institutions authenticate users through real-time video actions such as head movements or speech, and adversaries deploy face swaps or pre-recorded deepfake clips to create fraudulent accounts [30]. Industry projections quantify the exposure: Deloitte estimates that generative AI fraud, including deepfake attacks on financial systems, could cause up to $40B in losses by 2027 [5]. In response, organizations deploy deepfake detection models as automated gatekeepers in KYC pipelines. However, a critical question remains largely unaddressed: *do these detectors operate with equal reliability across the global customer base they are meant to serve?*

Demographic bias in deepfake detection manifests in two distinct perspectives (P1 and P2): **False-alarm disparities (P1)** (real faces mislabeled as fake): If a detector assigns systematically higher suspicion scores to real faces from certain demographic groups, then legitimate customers from those groups will experience higher false-alarm rates, leading to wrongful rejections, service disruptions, and erosion of trust in the service. In a regulated KYC context, this constitutes indirect discrimination and may violate equal-treatment legislation [8]. **Miss-rate disparities (P2)** (synthetic faces incorrectly classified as real): If a detector has higher miss rates for synthetic faces crafted with the appearance of particular demographic groups, then adversaries can selectively exploit these blind spots by targeting their attack personas to demographically underserved groups. This constitutes a security vulnerability with direct implications for fraud prevention.

Both failure modes are operationally relevant: over-detection harms inclusion and under-detection undermines security. Yet existing work overwhelmingly focuses on av-

erage aggregate performance, masking disparities that may affect millions of individuals in real deployment. In this study, we address three research questions (RQ1–RQ3).

RQ1 Do deepfake detectors reliably identify all generation categories?

RQ2 Do detectors exhibit systematic demographic bias in both false-alarm disparities on real faces (P1) and miss-rate disparities on synthetic faces (P2)?

RQ3 Can fine-tuning on demographically diverse data reduce these disparities?

To answer these questions, this paper makes four contributions (C1–C4):

C1 Multi-ethnic benchmark (FairFaceBench): a demographically labeled deepfake benchmark with self-reported ground-truth nationality and gender metadata, covering nine world regions and seven generation methods, yielding 35,424 labeled video frames.

C2 Dual-Perspective Bias Framework: a formal evaluation methodology that jointly examines false-alarm disparities (P 1) via per-group real-face scores and miss-rate disparities (P 2) via worst-group AUC (WG-AUC) and mean subgroup AUC deviation (ΔAUC) across demographic groups.

C3 Systematic empirical evaluation: we show that (a) no current detector reliably detects the newest avatar synthesis methods and (b) all evaluated detectors exhibit substantial and systematic cross-national bias.

C4 Bias Mitigation via Fine-Tuning: Fine-tuning on our demographically diverse benchmark dramatically reduces performance gaps and subgroup disparities, raising WG-AUC by 25-42 pp and halving ΔAUC. A worst-group loss (WGL) objective partially recovers the in-distribution (FF++) accuracy trade-off and yields additional fairness gains across generation categories, establishing a practical mitigation baseline.

2. Related Work

2.1. Deepfake Detection Models

RGB-Based Image Analysis: Early deepfake detection approaches mainly relied on training image classifiers on RGB input. For example, [1] proposed using Convolutional Neural Networks (CNNs) in combination with Error Level Analysis, achieving an accuracy of 91.83% on a self-collected dataset.

Biological Signal Features: Physiological signals such as rPPG can detect blood-flow artifacts in fake videos [13], with an AUC of 98% on Celeb-DF [17]. A broad overview of biological features and their effectiveness in deepfake detection is provided by [22]. [7] published a cautionary note on rPPG signals, highlighting that video compression can lead to unexpected results.

Frequency and Depth-Based Analysis: Recent methods also explore more subtle artifacts in fake content. Frequency-domain techniques have proven effectiveness in identifying inconsistencies that are hard to detect in the spatial domain [10].

Latent Space Feature Modeling: Another trend in deepfake detection involves analyzing latent features rather than pixel-level artifacts. The work of [3] introduces the StyleFlow model, which identifies manipulation patterns in the latent space of generative models.

We adopt three baseline detectors from DeepfakeBench [35], which unifies diverse open-source deepfake detection architectures under a common training and evaluation protocol, enabling fair and reproducible cross-method comparisons.

2.2. Fairness in Deepfake Detection

Demographic bias in face recognition has been documented extensively [2, 6, 29], but analogous analysis for deepfake detection is nascent.

Trinh and Liu [31] examine fairness in deepfake detection, evaluating three detectors across four ethnic groups (Caucasian, Asian, Indian, Black). They find that FaceForensics++, the dominant training corpus for deepfake detectors, is heavily skewed toward Caucasian subjects, with 36.6% of videos featuring female Caucasian faces alone. A further source of bias lies in the data generation pipeline itself: face-swap generation disproportionately produces cross-race and cross-gender composites, causing detectors to associate demographic appearance cues with fakeness rather than manipulation artifacts. When evaluated on gender- and race-balanced data, the studied classifiers exhibit predictive disparities of up to 10.7 percentage points in error rate across racial groups. Nadimpalli and Rattani [18] introduce a gender-balanced deepfake dataset but limit their analysis to male/female differences and lack ethnicity-level analysis.

Ju et al. [14] take an algorithmic approach to improve fairness in deepfake detection rather than merely measure it. They propose two loss-function wrappers based on Conditional Value-at-Risk: the first operates without demographic labels by up-weighting high-loss samples so that implicit minority groups receive lower error rates, whereas the second uses demographic labels and applies a hierarchical CVaR that equalizes per-group losses while separately handling the real/fake class imbalance within each group. Applied to five detectors across four datasets (FF++, Celeb-DF, DFD, DFDC), both methods reduce false-positive-rate gaps across gender and race groups without sacrificing overall AUC. However, the evaluation covers only four broad racial categories from existing benchmarks, none providing more

granular world-region level metadata or content from SOTA generative models. Their work motivated the adaptation of the loss function to up-weight minority groups during fine-tuning.

Xu et al. [33] provide a comprehensive bias audit, annotating five popular deepfake datasets with 47 demographic and non-demographic attributes (skin tone, accessories, etc.) yielding over 65 M labels. The ethnic granularity is Asian, Black, and White. Analyzing three detectors with a corrected relative performance metric that controls for test-set imbalance, they show that virtually all attributes influence detection accuracy, and crucially that the effect differs between real and fake images: a detector may simultaneously penalize a group on real faces and miss their fakes. This insight provided motivation to include P1 in our evaluation. While this work establishes the breadth of the bias problem, it proposes no mitigation and does not evaluate fine-granularity or cross-regional disparities. There is inherent uncertainty when automatically inferring demographic labels such as gender and ethnicity compared to relying on self-reported data, particularly from forged content.

The Fair Human-Centric Image Benchmark (FHIBE) [32], FairFace [15], and Casual Conversations [11] datasets were created to enable more balanced evaluation of face analysis systems but none is designed for deepfake detection. Leotta et al. [16] assemble an avatar deepfake benchmark using only open-source generative models, arguing that closed commercial systems cannot have their weights archived and that visible watermarks act as trivial detection shortcuts that bypass structural face-manipulation cues. We share this open-source motivation for our avatar category (InfiniteTalk) but extend the scope with richer ethnic coverage and by including the closed-source full-synthesis systems Veo 3.1 and Sora 2 as additional probes since both embed only steganographic watermarks in the API version with no directly visible artifact.

Research Gap. Taken together, existing fairness work in deepfake detection covers up to four broad racial groups [31, 14, 33], two genders [18], or automatically inferred attributes [33]. Avatar synthesis benchmarks exist [16, 34] but target detection performance rather than demographic bias and do not include world-region identity labels. No prior work combines all three: open-source avatar synthesis, commercial avatar synthesis, and verified nine-region demographic coverage in a unified bias evaluation. We address these gaps through our four contributions (C1–C4).

3. Dataset and Experimental Setup

3.1. Target Datasets

FHIBE [32] is a consent-based image dataset of 10,318 images from 1,981 unique individuals spanning 81 countries, with self-reported demographic labels at the identity level covering ancestry subregion, pronoun, age, and skin tone. FHIBE's GDPR compliance makes it a uniquely reliable source of verified-identity face images for fairness evaluation. We use FHIBE face crops as source identities in our video generation pipeline. They serve as conditioning input for avatar synthesis and face-swap methods (Section 3.2). For our analysis, we group FHIBE's country metadata into nine world regions to ensure large enough group sizes: *East/SE Asia, North America, Sub-Saharan Africa, Central/South Asia, S/E Europe, Latin America, W/N Europe, Oceania,* and *MENA*.

The identity-level annotation is the critical property for bias evaluation: all frames of a given subject carry consistent, verified demographic labels rather than labels inferred from facial appearance, which also avoids the implicit biases of human annotators or algorithmic bias [32]. This directly addresses the label ambiguity that makes demographic analysis unreliable in standard deepfake benchmarks.

We additionally use CelebV-HQ [37] as target material for the face-swap category: CelebV-HQ clips provide diverse real-world video contexts. For training the evaluated detectors, real frames are further supplemented with FairFace [15], a race-stratified static-image collection (seven ethnic backgrounds) that prevents the real-face training distribution from being dominated by any single region. FaceForensics++ [25] (FF++) is the standard in-distribution benchmark for deepfake detection, where the test split serves here as an in-distribution reference for measuring distribution-shift after fine-tuning.

Method	**Category**	**Mode**	**Duration hh:mm**
Gemini Veo 3.1 (2025) [9]	Commercial Avatar	T2V	00:49
Sora 2 (2025) [20]	Commercial Avatar	T2V	00:16
InfiniteTalk I2V (2025) [36]	ComfyUI Avatar	I2V	00:43
InfiniteTalk TTS (2025) [36]	ComfyUI Avatar	I2V + TTS	03:53
SimSwap (2020) [12]	Face swap	GAN encoder	01:42
HyperSwap (2023) [12]	Face swap	GAN + attn	01:22
InSwapper (2023) [12]	Face swap	GAN + ID loss	01:56
Total			**10:41**

Table 1. Generation methods by category. T2V = text-to-video (no face conditioning), I2V = image-to-video (FHIBE face as conditioning input), TTS = text-to-speech lip-sync. Face-swap methods replace the face in a CelebV-HQ target video with a FHIBE source identity.

3.2. Fake Content Generation Pipeline

We generated synthetic content across seven methods spanning three manipulation categories (Table 1). Each category produces qualitatively distinct artifact profiles, which are essential for separating generation-method difficulty from demographic bias in the analysis. Fig. 1 shows the resulting generated samples across all 18 region $\times$ gender

subgroups, confirming broad demographic coverage.

Avatar synthesis - Gemini Veo 3.1 and Sora 2. Both systems operate in pure text-to-video mode and no FHIBE face image is used as input since their content guardrails prevent generating fake videos of specific persons. For each target (nationality, gender, sentence) triplet, a structured prompt is assembled through a pipeline of deterministic lookups and random sentence content. The subject's sentence is drawn from a random sentence metafile which was generated using the OpenAI GPT-5.2 LLM. Diverse room types are also chosen (e.g., office, garages, ...), which are further described with atmosphere (e.g. garage $\rightarrow$ midday light or dim indoor, living room $\rightarrow$ also rainy). Clothing style is broadly conditioned (business / casual / creative) and then broken into per-category positive appearance slots, one option sampled per category (garment type, pattern, fit, color, accessory). All non-chosen options are written as explicit negative constraints to prevent the model from rendering them. The English speaking accent is sampled from a region-specific pool (e.g. Sub-Saharan Africa $\rightarrow$ West African accent), and age is drawn uniformly from six bins spanning 22-79 (22-29, then five 10-year bins). All combinations are injected into the structured prompt template together with a reference background image drawn from the Indoor Scenes Dataset [24]. The high diversity of backgrounds and also visual appearance should reduce the risk of short-cut learning. Sentences are in English, Sora 2 is generated at 1280×720, 8 s and Veo 3.1 at 16:9 aspect ratio at the API's native duration. Because no face image is used, the demographic label for full-synthesis content reflects the prompted identity, not a verified face. A manual spot-check confirmed the labels were reliable.

Open-Source Avatar synthesis - InfiniteTalk I2V and TTS. Both avatar methods take a FHIBE face crop (aligned, 768×768) as the conditioning image, so the demographic label is unambiguously that of the source subject. InfiniteTalk [36] I2V animates the face into a speaking video using WanVideo (wan2.1-i2v-14b, 23 denoising steps, CFG = 1.41, scheduler: DPM++ SDE). The fixed positive short prompt *"a person is speaking naturally"* and a standard negative prompt suppressing subtitles, distortion, and background motion ensure generation focuses on facial dynamics. InfiniteTalk TTS extends this pipeline with multilingual speech: a Qwen3-TTS model (1.7B parameters, 12 Hz vocoder) synthesizes random sentences in the subject's native language (e.g. French) using a voice instruction matching gender and regional accent (e.g. *"Male voice. Native Hindi speaker. Warm tone, natural Indian intonation"*). Wav2Vec-based audio embeddings drive the lip-sync functionality of WanVideo InfiniteTalk at 25 fps, 240 frames per clip (9.6 s). Both avatar variants are output in H.265 at CRF 22.

Face swaps - SimSwap, HyperSwap, InSwapper. All three face-swap variants share the same pairing and execution pipeline via FaceFusion [12], with the internal face-swap model selected per batch. The source face is always a FHIBE identity crop with self-reported and verified demographic labels. The target video is a CelebV-HQ clip, which carries no demographic annotations. We therefore applied the DeepFace classifier [27] to estimate the gender and race of the original person appearing in each CelebV-HQ clip. Source-target pairs were then matched so that the demographic appearance of the replacement face (FHIBE source) aligned with that of the original person (CelebV-HQ target), avoiding implausible composites such as a male face swapped onto a female body or a cross-racial substitution. A face-swapper blend weight of 1.0 is applied, meaning the face is visually dominated by the FHIBE source facial identity. The demographic label attaches to the source FHIBE identity, not the target video.

3.3. Evaluation Protocol and Generalization Design

Identity-safe train/val/test splits. For all videos traceable to a FHIBE subject (real FHIBE clips, avatar variants, face-swaps), we split at the subject identity level: no subject's face appears in both training and test partitions. Within each split, the assignment of subject identities is made as balanced as possible across gender and demographic strata.

For full-synthesis videos, where no FHIBE face is used, we split at the sentence level: every video sharing the same prompt sentence is assigned to the same partition. This prevents a detector from exploiting sentence-level content cues instead of visual artifacts when generalizing across generation methods. Fake videos are sampled at every 30th frame.

Leave-One-Method-Out Generalization. For cross-method generalization, we use leave-one-method-out (LOMO) evaluation across seven splits, each holding out one generation method. Detectors are fine-tuned on the remaining six and evaluated on the held-out set.

Artifact-shortcut prevention. All three face-swap methods and also commercial avatar videos produce baked-in black bars. Without correction, a detector can reach near-perfect face-swap accuracy by learning *"black bar $\Rightarrow$ fake"* rather than facial artifacts. We remove these bars with an automatic letterbox-stripping step.

3.4. Evaluated Detectors

We evaluate four frame-level classifiers: **DINOv2** [21] is a vision-transformer-based classifier from the Synthetiq Vision [19] V2 system, trained on a proprietary corpus

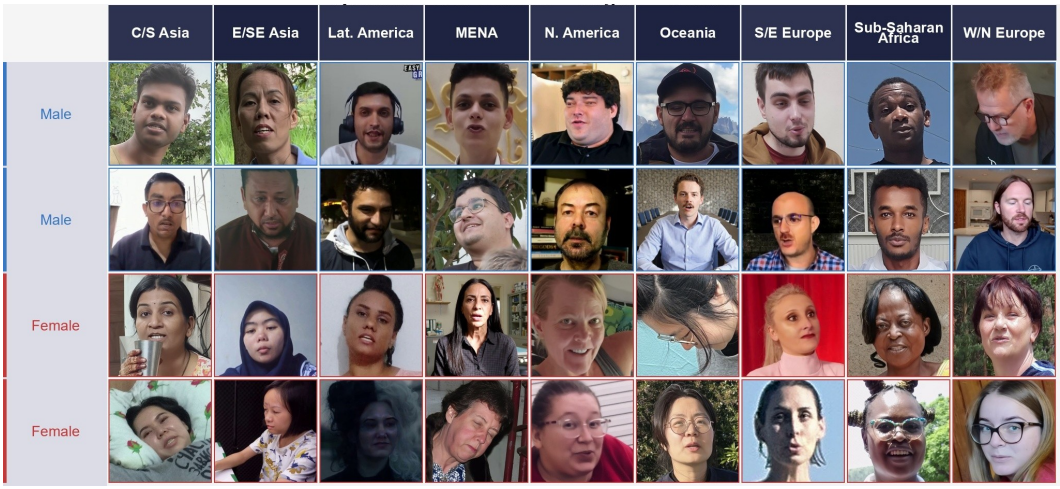


Figure 1. Generated samples across nine world regions (rows) and two genders (columns) in FairFaceBench.

with explicit upsampling for Asian subjects. The remaining three are standard DeepfakeBench [35] baselines trained on FF++ [25]: **EffNetB4** with an EfficientNet-B4 [28] backbone, **F3Net** with a frequency-domain CNN [23] backbone, and **Xception** [4].

All evaluation in this paper is conducted at the frame level. In production, video-level scores can be obtained by, for example, averaging per-frame scores within a clip.

3.5. Evaluation Framework

Here i indexes individual frames and all metrics are computed directly from frame-level scores without prior clip-level aggregation. Let $s_i \in [0, 1]$ denote the deepfake score assigned by a detector to frame i, where higher values indicate greater suspicion of being synthetic. Let $y_i \in \{0, 1\}$ denote the ground-truth label inherited from the clip containing frame i (0 = real, 1 = fake), and let $g_i \in \mathcal{G}$ denote the demographic group label (gender $\times$ world region) of the subject in that clip. We examine both failure modes using s_i: P1 via per-group mean detection scores on real faces (Section 4.1.2), and P2 via the two threshold-free metrics defined below.

Mean Subgroup AUC Deviation (ΔAUC) For synthetic videos ($y_i = 1$), let AUC_g be the area under the ROC curve computed with group g's fake frames as positives and a collection of FHIBE, CelebV-HQ, and FairFace frames as negatives, and let AUC denote the overall AUC across all groups. We measure average inter-group disparity as:

$$\Delta\mathrm{AUC} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \left| \mathrm{AUC}_g - \mathrm{AUC} \right|.$$

A higher $\Delta\mathrm{AUC}$ indicates that subgroup detection rates deviate more from the overall rate, signaling uneven demographic coverage.

Worst-Group AUC A higher AUC_g indicates the detector more reliably identifies fakes from group g, while a lower value signals a demographic blind spot exploitable by adversaries. We report:

$$\text{WG-AUC} = \min_{g \in \mathcal{G}} \mathrm{AUC}_g.$$

A low WG-AUC identifies the most vulnerable demographic group.

4. Results and Discussion

4.1. Bias Analysis

4.1.1 Overall Detector Performance

In-distribution (FF++) performance. All four detectors perform well on FF++ (EER 12-18%), confirming they are competent on the benchmark they were trained for.

Out-of-distribution breakdown. The aggregate "Overall" EER of 38-44% for the DeepfakeBench models (Table 2) is driven by two specific failure modes, not a general architectural limitation. First, commercial avatars (Veo 3.1, Sora 2) cause all four detectors to collapse: EER rises to 54-66% and mean score gaps turn negative, meaning the models score these fakes lower than real faces on average. Second, our newer face-swap implementations (SimSwap, HyperSwap, InSwapper) defeat the DeepfakeBench models (EER 40-45%) despite face swaps forming the core FF++ training data, indicating that their artifact profiles differ substantially from FF++ face swaps. However, for open-source avatars (InfiniteTalk), EffNetB4, F3Net, and Xception (EER 30-38%) match or outperform DINOv2 (EER 42%), suggesting that the frequency-domain and spatial artifacts produced by diffusion-based avatar synthesis are partially detectable by these architectures.

Subset	EER (%)↓	Mean gap (%)↑	$\bar{s}_{y=1}$ (%)↑	$\bar{s}_{y=0}$ (%)↓
DINOv2				
Overall	35.5 ± 1.00	31.5 ± 1.30	56.5 ± 1.10	25.0 ± 0.80
Face swap	16.7 ± 0.80	59.4 ± 1.20	84.4 ± 0.80	25.0 ± 0.90
Open-source avatar	42.0 ± 1.00	19.3 ± 1.70	44.3 ± 1.20	25.0 ± 0.90
Commercial av.	65.9 ± 0.90	−13.3 ± 1.00	11.7 ± 0.80	25.0 ± 0.90
FF++ in-dist.	11.9 ± 0.70	56.2 ± 1.20	93.7 ± 0.60	37.5 ± 1.00
EffNetB4				
Overall	44.3 ± 1.10	11.5 ± 1.20	42.8 ± 1.00	31.2 ± 0.70
Face swap	45.2 ± 0.90	8.1 ± 1.20	39.3 ± 0.90	31.2 ± 0.80
Open-source avatar	38.2 ± 1.00	21.6 ± 1.40	52.9 ± 1.00	31.2 ± 0.70
Commercial av.	64.6 ± 1.00	−12.8 ± 1.10	18.5 ± 0.80	31.2 ± 0.70
FF++ in-dist.	18.5 ± 0.80	56.6 ± 1.20	82.7 ± 0.70	26.1 ± 0.90
F3Net				
Overall	41.8 ± 1.00	16.7 ± 1.30	46.9 ± 1.00	30.2 ± 0.80
Face swap	43.6 ± 0.90	12.8 ± 1.30	43.1 ± 0.90	30.2 ± 0.80
Open-source avatar	35.1 ± 1.00	27.0 ± 1.30	57.2 ± 1.00	30.2 ± 0.90
Commercial av.	58.5 ± 1.00	−6.9 ± 1.20	23.4 ± 0.80	30.2 ± 0.90
FF++ in-dist.	17.5 ± 0.70	63.6 ± 1.20	81.0 ± 1.00	17.4 ± 0.80
Xception				
Overall	37.6 ± 1.00	18.6 ± 1.20	47.6 ± 1.00	29.0 ± 0.70
Face swap	39.9 ± 1.00	14.8 ± 1.00	43.9 ± 0.80	29.0 ± 0.60
Open-source avatar	30.6 ± 0.80	27.7 ± 1.00	56.7 ± 0.80	29.0 ± 0.60
Commercial av.	54.4 ± 0.90	−0.4 ± 1.00	28.6 ± 0.80	29.0 ± 0.60
FF++ in-dist.	18.3 ± 0.80	55.4 ± 1.10	83.7 ± 0.80	28.3 ± 0.80

Table 2. Detector performance by fake generation category and in-distribution FF++ baseline. ↑: higher is better; ↓: lower is better. : complete detector failure (EER > 50%, negative mean gap). EER = Equal Error Rate, Mean gap = $\bar{s}_{y=1} - \bar{s}_{y=0}$. ± values are 95% bootstrap CI half-widths (200 resamples, capped at 5,000 samples per class per iteration).

Region / Gender	DINOv2 (%)↑	EffNetB4 (%)↑	F3Net (%)↑	Xception (%)↑
W/N Europe ♀	55 ± 1	52 ± 2	49 ± 2	56 ± 2
W/N Europe ♂	46 ± 1	39 ± 1	36 ± 1	38 ± 1
Latin America ♀	57 ± 1	62 ± 2	63 ± 2	65 ± 2
Latin America ♂	52 ± 1	41 ± 2	39 ± 2	42 ± 2
MENA ♀	58 ± 3	47 ± 4	55 ± 4	52 ± 4
MENA ♂	36 ± 2	41 ± 2	52 ± 2	52 ± 2
C/S Asia ♀	53 ± 2	48 ± 3	51 ± 3	53 ± 3
C/S Asia ♂	73 ± 1	37 ± 1	44 ± 1	44 ± 1
Sub-Saharan Afr. ♀	67 ± 1	52 ± 1	62 ± 1	57 ± 1
Sub-Saharan Afr. ♂	61 ± 1	39 ± 1	52 ± 1	44 ± 1
East/SE Asia ♀	71 ± 1	52 ± 1	54 ± 1	59 ± 1
East/SE Asia ♂	57 ± 1	50 ± 2	60 ± 1	52 ± 2
Oceania ♀	64 ± 3	69 ± 4	68 ± 4	64 ± 4
Oceania ♂	26 ± 3	46 ± 5	59 ± 5	49 ± 5
S/E Europe ♀	60 ± 1	54 ± 2	53 ± 2	55 ± 2
S/E Europe ♂	54 ± 1	30 ± 1	32 ± 1	40 ± 1
North America ♀	68 ± 1	53 ± 2	50 ± 2	53 ± 2
North America ♂	65 ± 2	38 ± 2	42 ± 2	40 ± 2

Table 3. Zero-shot mean prediction score: the per-group mean of the raw detector score $s_i \in [0, 1]$ over fake frames (GT = 1), as a percentage. Higher = more likely to be flagged correctly. ↑: higher is better. ± values are Wilson 95% CI half-widths (rounded to 1 pp).
: lowest score per detector (worst-detected group).
: highest score per detector (best-detected group).

DINOv2 achieves the best overall EER of 35.5% and a mean score gap of 0.315, driven primarily by face-swap detection (EER 16.7%). It too fails on commercial-avatar content (EER 65.9%), confirming that no evaluated model weight set reliably identifies Veo or Sora output (**RQ1**).

4.1.2 False-Alarm Disparities on Real Faces (P1)

An unbiased deepfake detector would satisfy three conditions within each demographic subgroup: (i) real-face prediction scores should be consistent across genders; (ii) synthetic-face scores should likewise be consistent across genders; and (iii) a sufficient margin should separate real-face scores from synthetic-face scores. These conditions should hold uniformly across all world regions. As Fig. 2 shows, none of these conditions are fully met: within the MENA subgroup real-face scores differ substantially between male and female subjects (i), while for Oceania and W/N Europe the margin between real-face and synthetic-face scores is negligible (iii).

False-alarm pressure falls unequally: DINOv2 assigns systematically higher scores to real faces from female MENA and East/SE Asia identities than to others (Fig. 2), meaning those groups face disproportionate wrongful-rejection rates at any fixed operating threshold. Within all regions, female faces tend to attract higher false-alarm scores than male faces, making gender a consistent secondary axis of disparity. A similar pattern holds across the DeepfakeBench models, indicating this is a data-driven bias rather than an architectural quirk.

4.1.3 Miss-Rate Disparities on Synthetic Faces (P2)

Miss-rate disparities mirror the false-alarm pattern. DINOv2 detection scores on forged content range from 0.26 (Oceania *m*) to 0.73 (C/S Asia *m*), a gap that translates into an exploitable blind spot for targeted fraud (Table 3). The high scores for C/S Asian males align with DINOv2's proprietary upsampling of Asian subjects, illustrating targeted data curation can close specific blind spots. W/N and S/E European males are among the groups most represented in standard training benchmarks [31], yet they receive weak detection scores. Together, P1 and P2 identify largely non-overlapping at-risk populations. False-alarm disparities burden female MENA and Asian identities, while miss-rate blind spots leave male European fakes most exploitable (**RQ2**).

4.2. Fine-Tuning for Demographic Robustness

The bias patterns quantified in Section 4 partially arise from detectors trained on demographically narrow benchmarks and lack of generalization [33]. We investigate whether fine-tuning on FairFaceBench can reduce these disparities, and introduce a WGL objective to target the hardest demographic groups during training.

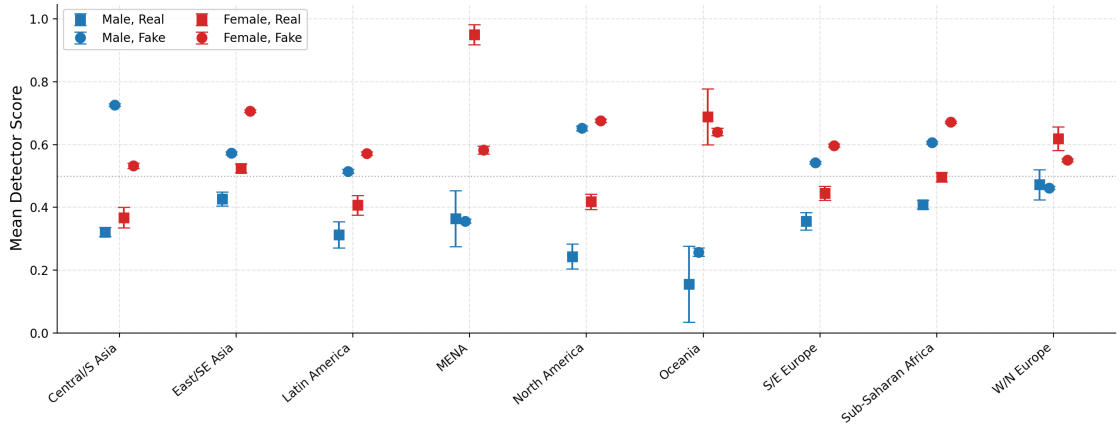


Figure 2. Mean DINOv2 prediction score by world region, gender, and ground-truth label (real = GT 0, fake = GT 1), with ±1 SEM error bars. Higher GT = 0 scores indicate elevated false-alarm pressure on real faces (P 1); lower GT = 1 scores indicate harder-to-detect fakes (P 2).

4.2.1 Detector Architectures and Fine-Tuning Setup

The Synthetiq Vision [19] model uses a DINOv2 ViT-L/14 [21] backbone trained on a proprietary dataset. For fine-tuning, we adopt partial unfreezing: the patch embedding, positional embedding, and the first $(L - k)$ transformer blocks are frozen. Only the last k blocks, the CLS token, and the linear head receive gradient updates. This preserves the rich general visual representations while adapting the decision boundary to our demographically rich dataset distribution. We treat k as a hyperparameter (HP). DeepfakeBench [35] is a benchmark unifying multiple backbone models and also datasets. We choose the following models since their weights are provided in the repository: EffNetB4 uses compound-scaled inverted residual blocks, Xception replaces standard convolutions with depthwise separable convolutions, reducing parameter count while maintaining receptive field and F3Net augments a spatial CNN with learnable frequency-domain filters that explicitly decompose the face into multi-scale frequency components, targeting the spectral artifacts left by GAN upsampling.

All three are trained end-to-end from pre-trained DeepfakeBench weights. For fine-tuning we unfreeze all parameters, consistent with standard DeepfakeBench practice.

4.2.2 Training Protocol

Fine-tuning follows the LOMO evaluation protocol described in Section 3.3: for each held-out generation method T, we fine-tune a pre-trained checkpoint on all remaining fake types and evaluate on T. We conduct a small grid search over the learning rate, unfrozen blocks k for DINOv2, LR schedule and epochs, selecting the configuration with the best mean validation AUC across all LOMO folds (validation uses a held-out split of T, disjoint from the test videos).

4.2.3 Worst-Group Loss for Demographic Debiasing

Standard gradient descent minimizes average loss, often allowing majority groups to dominate the gradient and degrading minority-group performance [26]. In deepfake detection, this can skew decision boundaries, leaving minority-group fakes systematically harder to detect. Worst-Group Loss (WGL) addresses this by explicitly penalizing the worst-performing group [26]. Let $\mathcal{G}$ denote the set of demographic groups, defined as the Cartesian product of gender ($\in \{female, male\}$) and world region ($\in 9$ regions), giving $|\mathcal{G}| = 18$ groups. Let $D_g = \{i : g_i = g\}$ be the set of training samples belonging to group g, and let $\ell(f_\theta(x_i), y_i)$ denote the binary cross-entropy loss for sample i. The per-group loss is:

$$L_g(\theta) = \frac{1}{|D_g|} \sum_{i \in D_g} \ell(f_\theta(x_i),\, y_i)\,.$$

We optimize a convex combination of the average loss and the WGL:

$$L(\theta) = (1-\alpha) \underbrace{\frac{1}{N} \sum_{i=1}^{N} \ell(f_\theta(x_i),\, y_i)}_{L_{avg}(\theta)} + \alpha \underbrace{\max_{g \in \mathcal{G}} L_g(\theta)}_{L_{\mathrm{WGL}}(\theta)},$$

where $\alpha \in [0, 1]$ is a tuned coefficient controlling the fairness-accuracy trade-off. Final HPs are listed in Table 4.

Model	α	LR	Sched.	Unfrz. k
DINOv2	0.20	1.0×10^{-4}	-	8
Xception	0.25	3.0×10^{-5}	-	-
EfficientNetB4	0.25	4.3×10^{-5}	-	-
F3Net	0.37	3.0×10^{-4}	cosine	-

Table 4. Selected HPs (best mean val. AUC across LOMO folds).

4.2.4 Fine-Tuning Results

As Table 5 shows, all four zero-shot baselines exhibit moderate overall AUC (58-71%) but severe worst-group bias (WG-AUC 41-51%). Fine-tuning (FT) on our diverse training split yields overall AUC gains of +19 to +26 pp, raises WG-AUC by 25-42 pp, and roughly halves ΔAUC, proving the value of demographic training coverage (**RQ3**).

Commercial Avatar AUC after FT reaches 88-97% which is the highest of all generation categories, despite being the hardest category at baseline. We hypothesize that the fine-tuned models exploit superficial generation cues such as background patterns, scene-level lighting inconsistencies, or spectral artifacts introduced by full-scene diffusion synthesis. Under LOMO, when one commercial system (e.g. Sora 2) is held out for evaluation, the model is still exposed to the other (Veo 3.1) during training. Since both are full-scene diffusion systems, they share characteristic spectral signatures, allowing the detector to transfer its learned artifact representation. The results align with [16], who also note strong post-fine-tuning avatar detection but reduced in-distribution accuracy. Performance gains come at the cost of lower FF++ in-distribution AUC (7-24 pp drop), suggesting catastrophic forgetting as models adapt to artifacts in our training split. Adding WGL partially recovers this penalty: FF++ AUC is 1-2 pp higher than standard FT for DINOv2 and Xception, and yields additional fairness gains, most notably for DINOv2 Face Swap (WG-AUC 79.3 % $\rightarrow$ 88.7 %), while overall AUC and ΔAUC remain comparable to FT across architectures. After FT + WGL, the hardest subgroup shifts to ♂/E. Asia (DINOv2, EfficientNetB4) and ♂/S. Europe (Xception, F3Net), reflecting residual dataset imbalances.

5. Conclusion

We introduced FairFaceBench, a demographically controlled deepfake benchmark spanning nine world regions, two genders, and seven generation methods, and used it to audit four widely used detectors from two bias perspectives: false-alarm disparities on real faces and miss-rate disparities on synthetic faces. Zero-shot evaluation reveals that all tested detectors are evaded by the newest avatar synthesis systems (**RQ1**) and exhibit demographic bias (WG-AUC 41-51%, ΔAUC up to 9.7 pp) (**RQ2**). Fine-tuning narrows the gap (overall AUC +19-26 pp, WG-AUC +25-42 pp, ΔAUC halved), at the cost of a distribution-shift penalty on FF++ (7-24 pp AUC drop) that a WGL objective partially recovers while adding further fairness gains (**RQ3**). While diverse, FairFaceBench is finite with some wide subgroup confidence intervals, and future synthesis methods will inevitably evade current models. We recommend that practitioners evaluate on demographic benchmarks and consider fine-tuning to reduce bias, train detectors on diverse datasets, and maximize demographic coverage when constructing deepfake datasets.

Method / Domain	AUC↑	WG-AUC↑	Worst Group	ΔAUC↓
DINOv2				
Baseline: Overall	70.60 ±1.02	41.22 ±3.01	♂/Oce.	7.77 ±0.48
Face Swap	91.13 ±0.58	75.27 ±2.67	♀/W.Eu.	3.96 ±0.42
Open-Src Avatar	62.40 ±1.09	38.24 ±2.90	♂/Oce.	8.96 ±0.49
Comm. Avatar	27.78 ±1.18	3.60 ±0.88	♂/SSA	15.71 ±2.06
FF++	94.51 ±0.46	-	-	-
FT: Overall	90.39 ±0.62	83.40 ±0.82	♂/S.Eu.	2.34 ±0.23
Face Swap	90.39 ±0.56	79.30 ±4.01	♂/Oce.	3.28 ±0.31
Open-Src Avatar	88.89 ±0.62	76.96 ±1.15	♂/S.Eu.	3.80 ±0.27
Comm. Avatar	96.93 ±0.26	89.36 ±6.79	♂/Oce.	2.07 ±0.55
FF++	86.77 ±0.69	-	-	-
FT + WGL: Overall	89.86 ±0.63	80.57 ±2.89	♂/E.Asia	2.56 ±0.28
Face Swap	92.75 ±0.51	88.66 ±1.98	♂/Oce.	2.93 ±0.28
Open-Src Avatar	85.12 ±0.80	71.14 ±1.24	♂/S.Eu.	3.80 ±0.38
Comm. Avatar	95.92 ±0.33	91.50 ±3.61	♂/Oce.	3.08 ±0.53
FF++	89.09 ±0.63	-	-	-
Xception				
Baseline: Overall	66.74 ±1.10	51.28 ±1.47	♂/W.Eu.	8.33 ±0.58
Face Swap	62.79 ±1.00	34.94 ±1.44	♂/W.Eu.	10.96 ±0.57
Open-Src Avatar	74.57 ±1.03	63.29 ±4.77	♂/E.Asia	7.95 ±0.66
Comm. Avatar	52.61 ±1.23	23.61 ±4.26	♀/SSA	22.97 ±4.25
FF++	89.83 ±0.64	-	-	-
FT: Overall	87.49 ±0.71	82.06 ±0.87	♂/S.Eu.	2.57 ±0.21
Face Swap	86.61 ±0.71	82.36 ±0.90	♂/E.Asia	2.75 ±0.23
Open-Src Avatar	87.25 ±0.67	79.32 ±0.97	♂/S.Eu.	3.16 ±0.30
Comm. Avatar	92.06 ±0.51	78.96 ±4.98	♂/MENA	4.52 ±0.78
FF++	75.87 ±1.00	-	-	-
FT + WGL: Overall	83.02 ±0.73	76.02 ±1.43	♂/Oce.	3.11 ±0.24
Face Swap	82.89 ±0.78	77.05 ±1.55	♂/E.Asia	4.04 ±0.26
Open-Src Avatar	82.12 ±0.78	74.76 ±1.02	♂/S.Eu.	3.83 ±0.32
Comm. Avatar	86.72 ±0.69	65.44 ±4.49	♂/Oce.	6.81 ±0.88
FF++	77.14 ±0.91	-	-	-
EfficientNetB4				
Baseline: Overall	57.63 ±1.05	44.59 ±2.35	♂/W.Eu.	9.53 ±0.66
Face Swap	55.94 ±1.16	31.27 ±1.50	♂/W.Eu.	11.79 ±0.61
Open-Src Avatar	64.55 ±1.04	46.77 ±4.55	♂/S.Asia	8.84 ±0.85
Comm. Avatar	35.52 ±1.26	9.36 ±3.20	♂/SSA	19.47 ±3.47
FF++	89.82 ±0.60	-	-	-
FT: Overall	83.55 ±0.80	74.97 ±1.37	♂/E.Asia	3.47 ±0.28
Face Swap	81.61 ±0.75	71.33 ±1.27	♂/E.Asia	5.80 ±0.31
Open-Src Avatar	84.45 ±0.76	73.91 ±1.13	♂/S.Eu.	3.12 ±0.33
Comm. Avatar	88.62 ±0.66	87.60 ±5.06	♀/Lat.Am.	4.32 ±0.85
FF++	70.33 ±1.13	-	-	-
FT + WGL: Overall	80.15 ±0.82	71.98 ±1.08	♂/E.Asia	3.36 ±0.29
Face Swap	78.90 ±0.88	67.98 ±2.17	♂/Oce.	5.99 ±0.35
Open-Src Avatar	80.51 ±0.79	73.97 ±1.89	♂/S.Eu.	2.99 ±0.34
Comm. Avatar	84.17 ±0.79	78.82 ±3.55	♂/Oce.	5.46 ±0.80
FF++	68.41 ±1.07	-	-	-
F3Net				
Baseline: Overall	62.95 ±1.04	49.11 ±1.51	♂/W.Eu.	9.68 ±0.64
Face Swap	59.75 ±1.07	35.28 ±1.51	♂/W.Eu.	11.94 ±0.64
Open-Src Avatar	70.61 ±1.02	56.12 ±3.45	♂/S.Asia	8.88 ±0.83
Comm. Avatar	45.54 ±1.31	18.07 ±3.65	♂/SSA	21.38 ±3.77
FF++	90.50 ±0.63	-	-	-
FT: Overall	88.23 ±0.66	79.45 ±0.93	♂/S.Eu.	2.57 ±0.25
Face Swap	86.74 ±0.69	72.26 ±3.48	♀/MENA	4.63 ±0.35
Open-Src Avatar	88.95 ±0.60	79.37 ±1.03	♂/S.Eu.	2.76 ±0.23
Comm. Avatar	91.97 ±0.53	77.78 ±5.13	♂/Oce.	3.59 ±0.75
FF++	66.21 ±1.01	-	-	-
FT + WGL: Overall	87.35 ±0.71	79.18 ±0.94	♂/S.Eu.	2.73 ±0.25
Face Swap	87.12 ±0.67	69.45 ±3.09	♂/Oce.	5.36 ±0.30
Open-Src Avatar	86.46 ±0.77	76.48 ±1.16	♂/S.Eu.	3.27 ±0.31
Comm. Avatar	91.49 ±0.61	76.03 ±4.76	♂/Oce.	4.08 ±0.72
FF++	66.33 ±0.98	-	-	-

Table 5. Fine-tuning results on the test split. ↑: higher is better; ↓: lower is better.
: failure or active harm (reduction compared to baseline AUC; Fine-Tuning degrading WG-AUC).
: improvement over baseline (peak fairness).
WG-AUC: worst-group AUC (minimum across all gender × region subgroups). ΔAUC: mean $|\text{AUC}_{\text{group}} - \text{AUC}_{\text{overall}}|$ (lower = more uniform subgroup coverage). All values: point estimate ± 95% bootstrap CI (half-width, n=500).

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